

PATARI v0.2 Instructions

Analysis Tool for Clinical Photoacoustic Imaging

<https://github.com/Mo-Sc/PATARI>

UNTESTED VERSION – USE ONLY FOR EVALUATION PURPOSES

-
- Description
 - Features
 - Installation
 - Usage
 - Example Workflow

-
- An open source GUI tool that covers the most common analysis workflows for clinical photoacoustic studies
 - Target group: medical doctors / non-technical users
 - Pure-python based
 - Runs on Windows / MacOS / Linux
 - No additional dependencies, easy installation
 - Relies on established frameworks:
 - NAPARI (GUI)
 - PATATO (Data I/O, Processing)
 - Modular architecture:
 - Can be adapted to individual needs
 - As simple as possible, as complex as necessary

Implemented or planned

- Supports different file formats + easy extension to new ones
 - Native iThera (including iLabs annotations)
 - PATATO HDF5
 - IPASC (not included in v0.2)
 - SIMPA (not included in v0.2)
- Seamless scrolling through frames and wavelengths
- Overlay / side by side view of US and PA images
- Overlay / side by side view of different reconstructions
- Spectral unmixing (provided by PATATO)
- Postprocessing / reconstruction (provided by PATATO) (not included in v0.2)
- Export to HDF5
- **Quantitative image analysis:**
 - ROI placement and intensity extraction using NAPARI shapes
 - Import / export of ROIs + customizable default ROIs, persistent ROI-library
 - Auto-updating live ROI table / Persistent saved ROI table
 - Plot PA intensity over time for multiple ROIs
 - Histogram view for multiple ROIs
 - Plot ROI wavelength / chromophore spectra
 - Analysis export to Microsoft Excel
- **Automation:** (not included in v0.2)
 - Deep learning based tissue segmentation from US image (<https://doi.org/10.1117/12.3049067>)
 - Automatic ROI placement in target class (i.e. muscle, fat layer, ...)
 - Batch processing mode

full upcoming feature list on [github](#)

1. Install the **napari** app:

- https://napari.org/stable/getting_started/installation.html#installation-bundle-conda
- Select the installer that corresponds to your OS (Mac / Windows) and follow the instructions
- Launch **napari** from launchpad / start menu

2. Install the **PATARI** plugin:

- In **napari**, click **Plugins** → **Install/Uninstall Plugins**
- Drag and drop the included .whl file into the plugin manager window
- Next to the Install button, select PyPI and click Install
- After installation is done, restart napari, and select PATARI Controls under Plugins

Usage

Overview (widget positions can differ)

Layer Settings Dock

Image layers: control contrast, opacity, colormap

ROI Layer: Manually define ROIs to place in scan

Layer List

List of all layers included in the scan + ROI layer

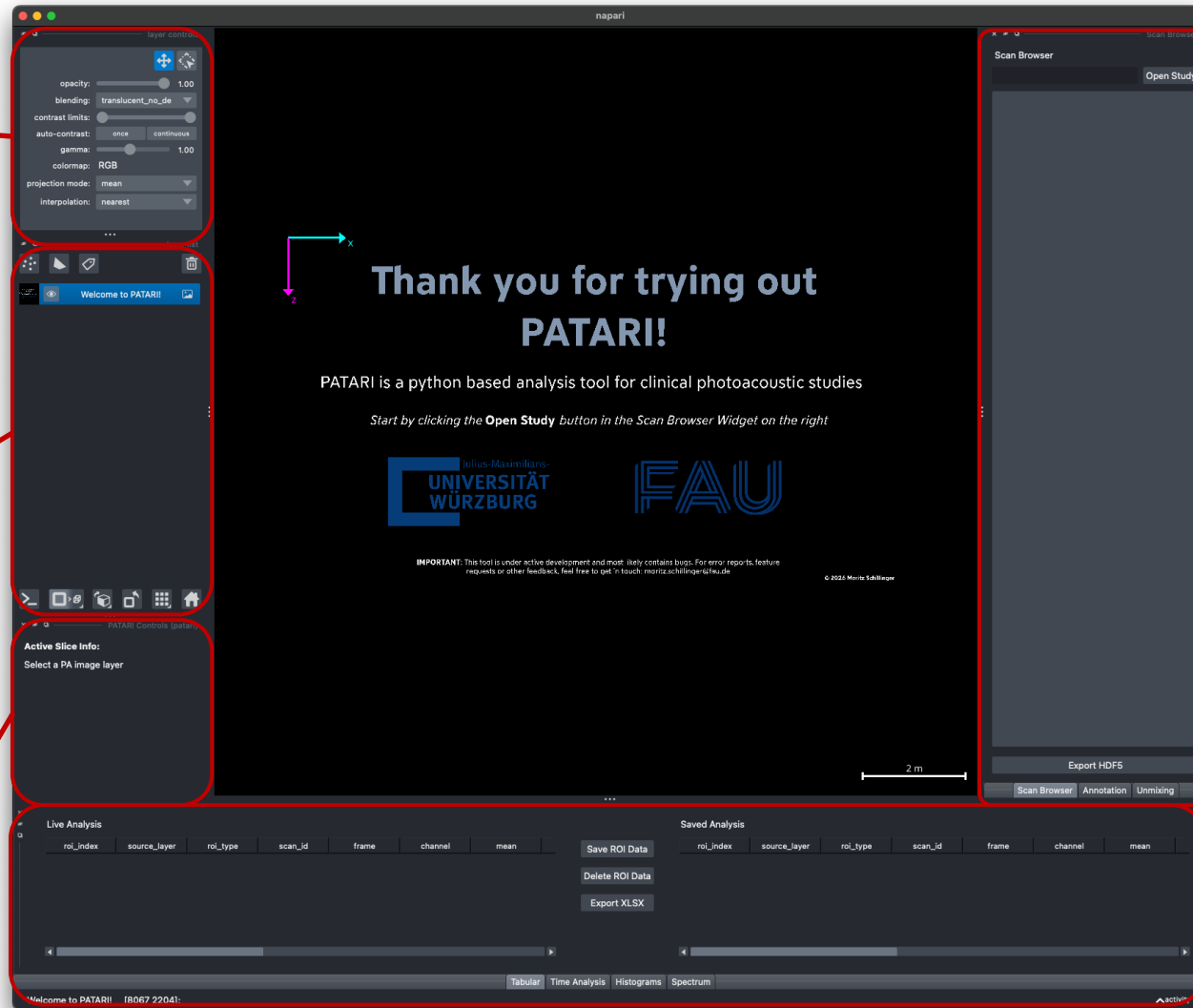
Selecting a PA makes it the **Active Layer**. All other PA layers are automatically set to hidden (eye button)

Processing and ROI analysis is always performed on the **Active Layer**

Slice-Info Dock

Shows relevant information about the selected slice

A slice refers to a single 2D image of a given layer, frame and channel (wavelength or chromophore)



Control Dock

Scan Browser: Open Studies, switch Scans, export scans to HDF5

Annotation: ROI-Library, ROI settings

Unmixing: Create chromophore layers

ROI-Analysis Dock

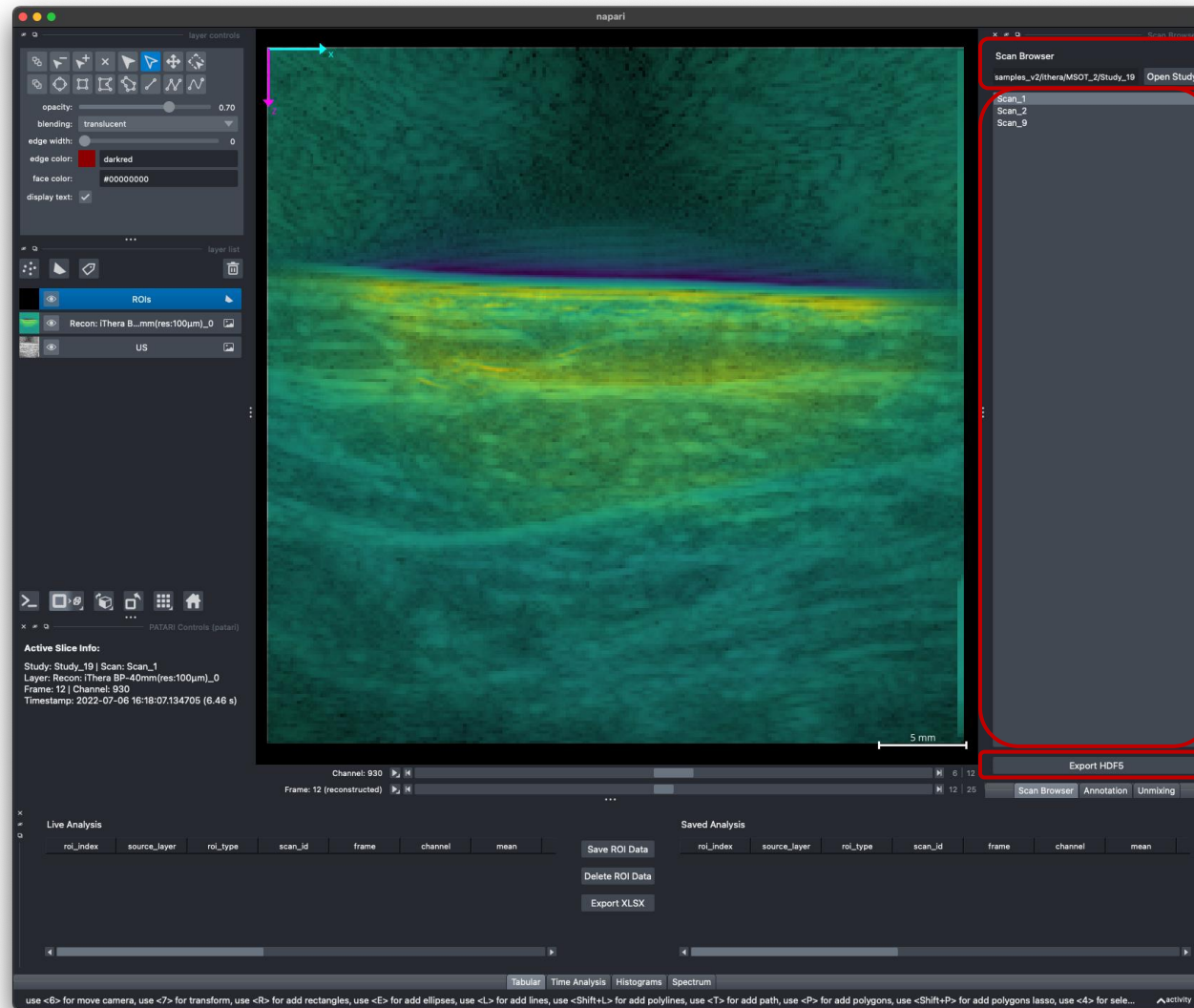
Extracts the data within the ROIs from the **Active Layer**

Tabular: Calculates and collects statistical features in the ROIs

Time Analysis: Displays mean intensity in the ROIs over time

Histogram: Shows histogram of ROI intensities at selected frame and channel

Spectrum: Shows ROI intensities over channel (wavelength or chromophore)



Open study folder or individual scan folder

If multiple scans are present in the study folder, they will be listed here

By default, first scan will be opened

Switch to other scan by clicking on the scan name

For large scans, opening might take a few seconds

Exports the opened scan into a HDF5 file (see [here](#) for more info)

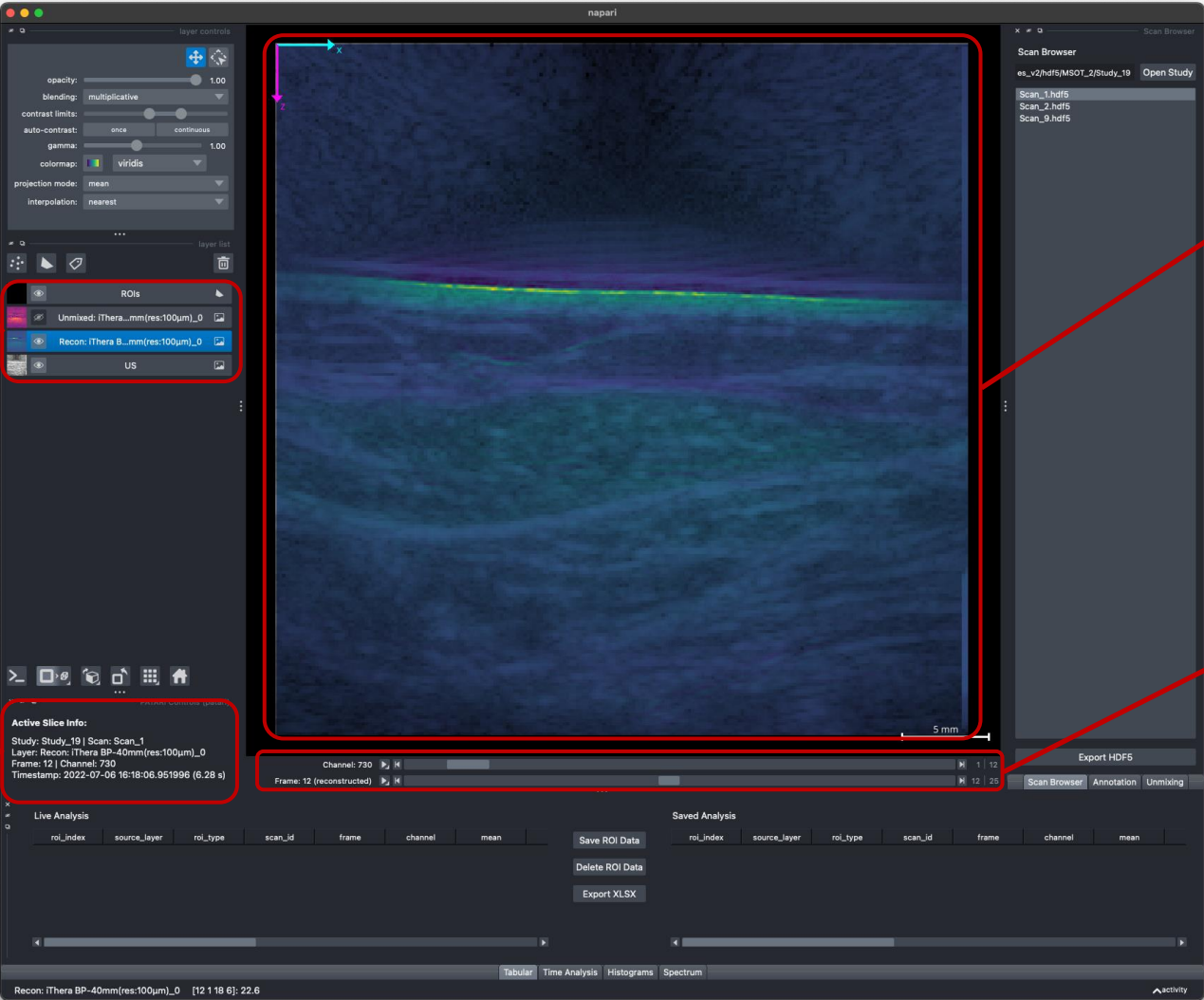
Includes all data generated by the plugin (currently ROIs and unmixed datasets)

Exported HDF5 files can be opened using this plugin again

Overwrite policy:

- Data always has to be exported to a new file (cant append to existing HDF5 file)
- Data that was in the original file **always persists**
- Data that was previously generated by PATARI is **always overwritten**

Image Viewer



This layer list contains two PA layers (Recon & Unmixed). The Recon layer is selected as active layer and will therefore be used for processing and analysis.

The Slice-Info Dock also shows the active layer, selected frame and channel, as well as the timestamp (if available).

Main Image Viewer

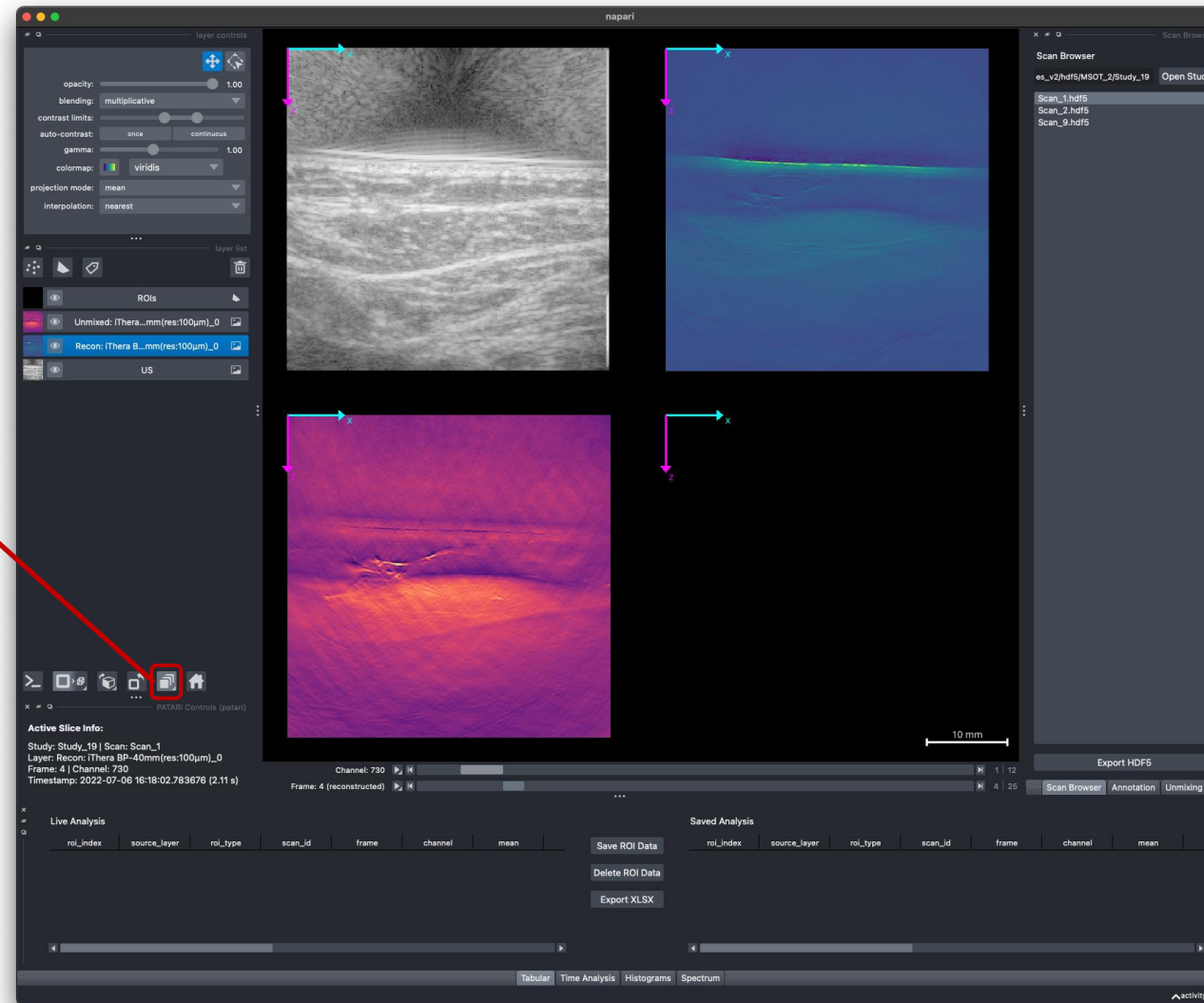
By default, the viewer shows the ultrasound layer, overlaid with a reconstructed PA layer, and the ROI layer. This can be changed in the layer list and the layer settings.

The two sliders below the viewer are used to select the frame and the channel. Channel can be either wavelengths or chromophores. The indices are synced across all layers. The available indices and the slider labels depend on the active layer.

Grid Mode

Clicking on the grid mode button displays the layers next to each other instead of an overlay.

Note that the ROIs layer is shown as a separate grid element. Grid mode is therefore not useful for annotation (napari limitation).



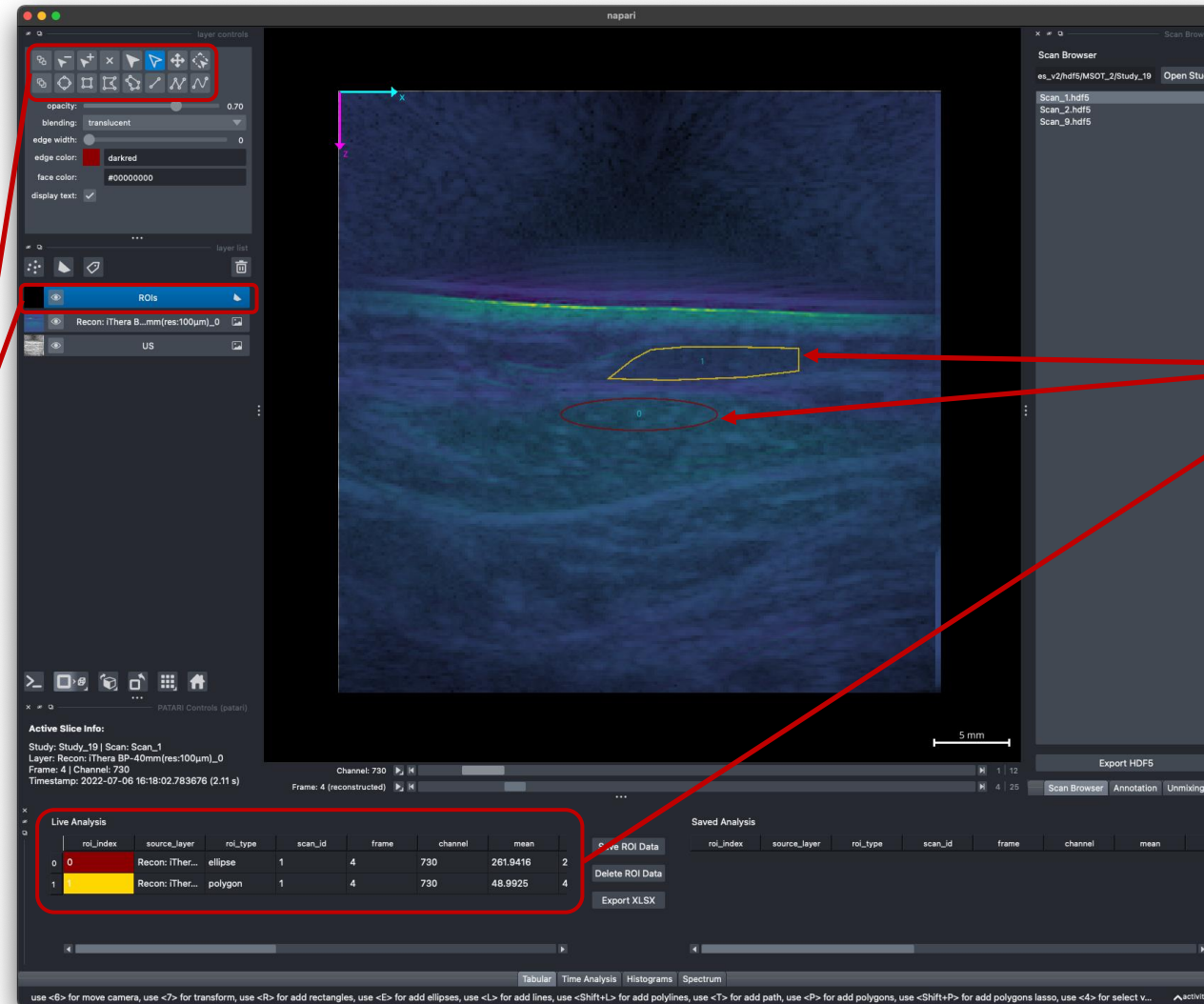
Manual ROI Drawing

To start drawing ROIs, select the **ROIs layer** in the layer list.

This will show the ROI layer controls in the layer settings:

- Select  or press **E** to draw an ellipse
- Select  or press **R** to draw a rectangle
- Select  or press **P** to draw an ellipse
- Select  or press **S** to select, move, or resize an existing ROI
- Select  or press **D** to edit polygon ROIs
- Select  or press **DEL** to delete the selected ROI

Note: They keyboard shortcuts only work when the ROIs layer is selected



New ROIs are automatically added to the **Live Analysis Table**.

To remove a ROI from the table, delete it in the viewer.

ROIs can be identified by **ROI ID** and **colour**. The ROI features in the Live Analysis Table are **automatically updated** when switching between active layers or scrolling through wavelengths / channels.

To select a ROI, select the ROIs Layer and use the select tool (shortcut **S**) and click on the ROI. Newly drawn ROIs are automatically selected.

Live Analysis Table

This table lists all the ROIs that are currently drawn in the viewer.

Imported ROIs (i.e. created by iLabs and included in the file) are also displayed here.

The extracted ROI data auto-updates when layer, frame or channel change.

ROIs are scan-specific, and are dropped when a new scan is opened.

The ROIs in the Live Table are written to file, when the scan is exported using the Export HDF5 button in the scan browser.

Save ROI Data: Copies the selected ROI (here ID 1) from the Live Table to the Saved Table

Delete ROI Data: Removes the selected row(s) from the Saved Table

Export XLSX: Exports the Saved Table into a MS Excel file.

Saved Analysis Table

Once a Live Table entry was added to the Saved Analysis Table, it does not change anymore and is saved, even when a new scan or study is opened.

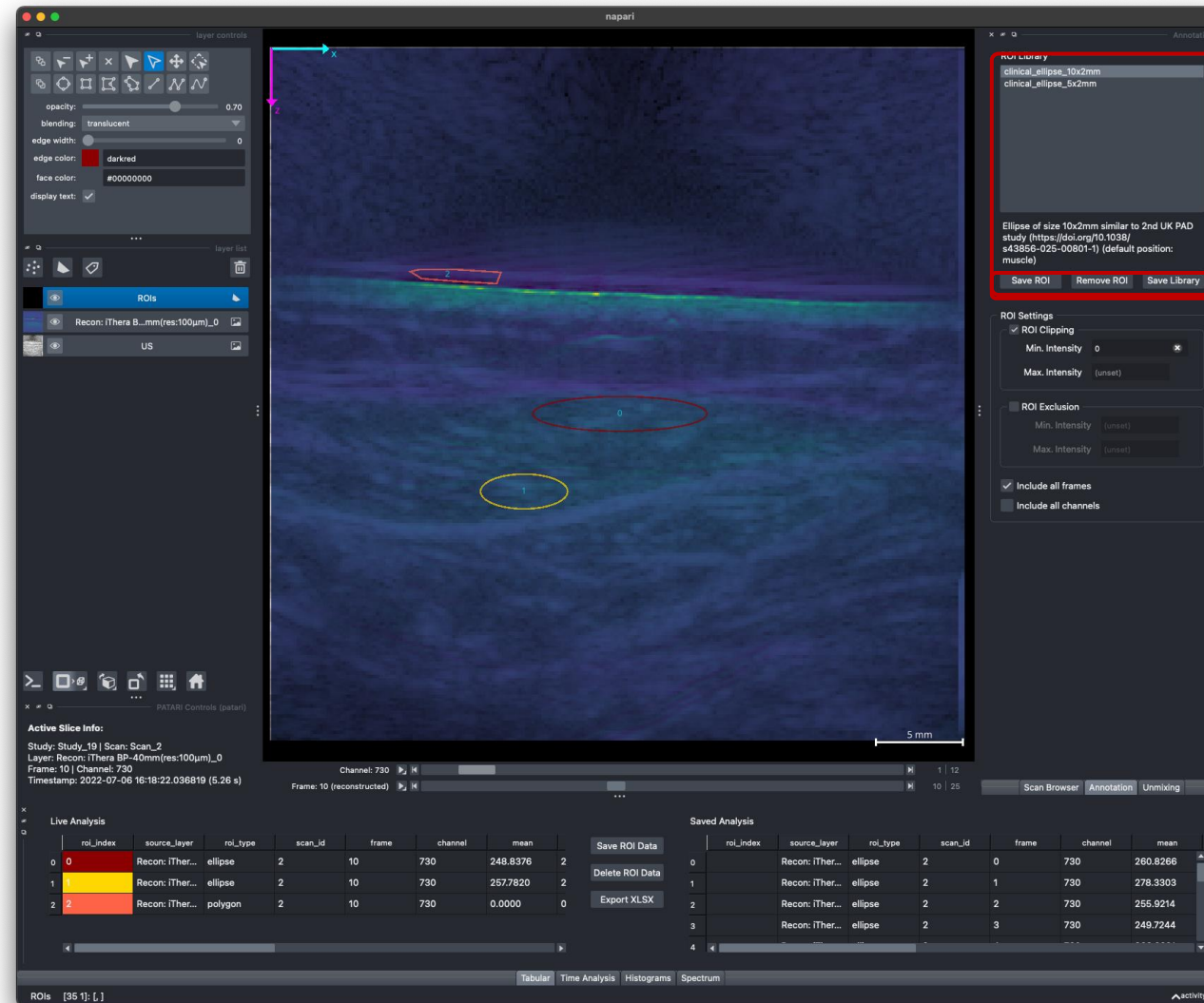
Example: 3 ROIs were saved. One ROI (3rd row) was added from the current Live Analysis Table. But the other two were saved from a different wavelength (2nd row) or even a different scan (1st row).

The screenshot shows the napari software interface. The central panel displays a 3D volume rendering of a sample with several regions of interest (ROIs) highlighted in different colors (yellow, red, green, blue). The left sidebar contains the 'layer controls' and 'layer list' panels. The bottom panel shows the 'Live Analysis' and 'Saved Analysis' tables. The 'Live Analysis' table has 5 rows, and the 'Saved Analysis' table has 3 rows. The 'Export HDF5' button is visible in the bottom right corner of the interface.

roi_index	source_layer	roi_type	scan_id	frame	channel	mean	
0	Recon: ITher...	polygon	9	6	700	422.3687	4
1	Recon: ITher...	ellipse	9	6	700	235.0433	2
2	Recon: ITher...	ellipse	9	6	700	215.8244	2
3	Recon: ITher...	ellipse	9	6	700	186.2607	1
4	Recon: ITher...	ellipse	9	6	700	-59.9938	-

roi_index	source_layer	roi_type	scan_id	frame	channel	mean	
0	Recon: ITher...	polygon	1	12	930	394.9211	3
1	Recon: ITher...	polygon	9	12	760	391.5608	3
2	Recon: ITher...	ellipse	9	6	700	235.0433	2

Annotation Widget – ROI Library



ROI Library

Contains preconfigured ROI shapes.

Can be used to quickly draw identical ROIs across multiple studies and scans.

Clicking on a ROI will show its description.

Double-clicking will place the ROI into the viewer (and add it to the Live Table).

ROIs can be added or removed to/from this library.

Added entries will persist as long as the app runs.

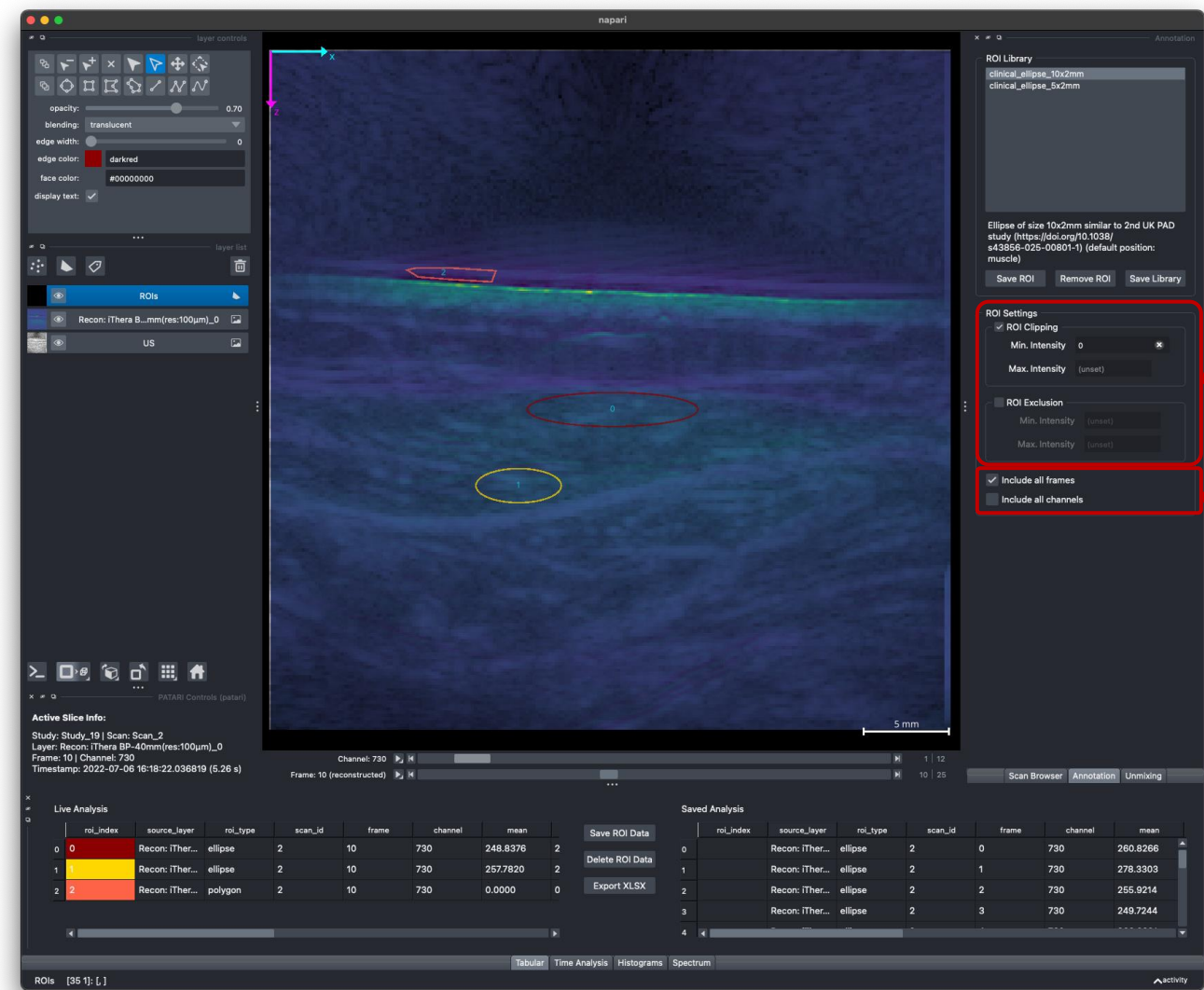
Library is connected to a ROI file, so ROIs can be shared across devices.

Save ROI: Adds the ROI shape that is selected in the viewer to the library.

Remove ROI: Removes the ROI that is selected in the library from the library.

Save Library: Permanently saves the ROI Library to file so it persists even when the app is closed.

Annotation Widget – ROI Settings



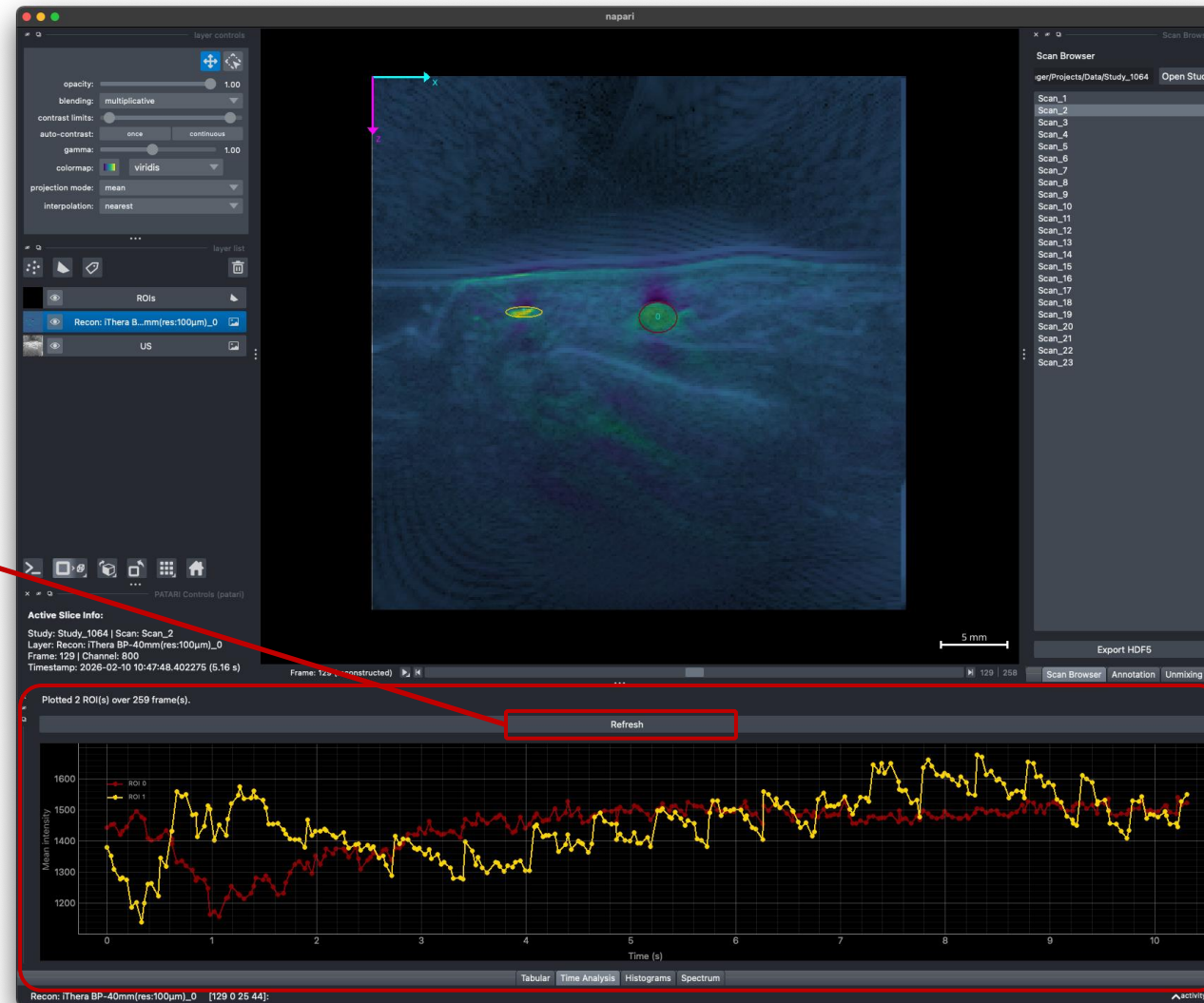
Settings to control which intensities are included in the ROI feature calculations. Settings are automatically applied to all the ROIs in the Live Table

ROI Clipping: Clips all the intensities in the ROIs to the given range

ROI Exclusion: Excludes all the intensities outside the given range from the feature calculation

Controls the behaviour of the **Save ROI Data** button

If checked, saving an ROI will add entries for all the frames and/or channels of the active layer to the Saved Table, instead of just the selected slice



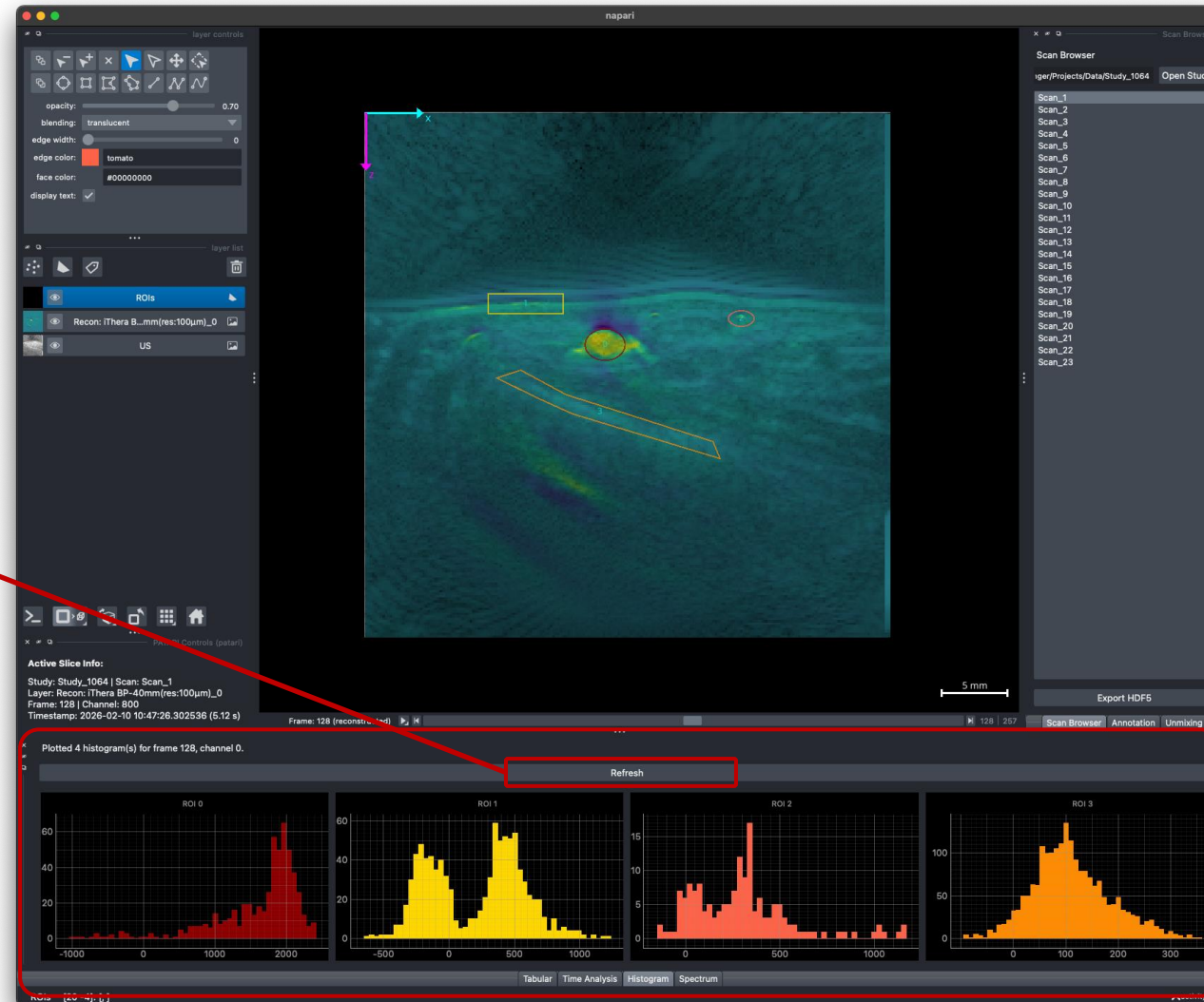
Unlike the Live Table, plots are not auto-updated. When ROIs, layer or channel are changed, **Refresh** has to be clicked again.

Plots ROI intensities over time.

y-axis: mean intensity in the ROI for the selected channel.

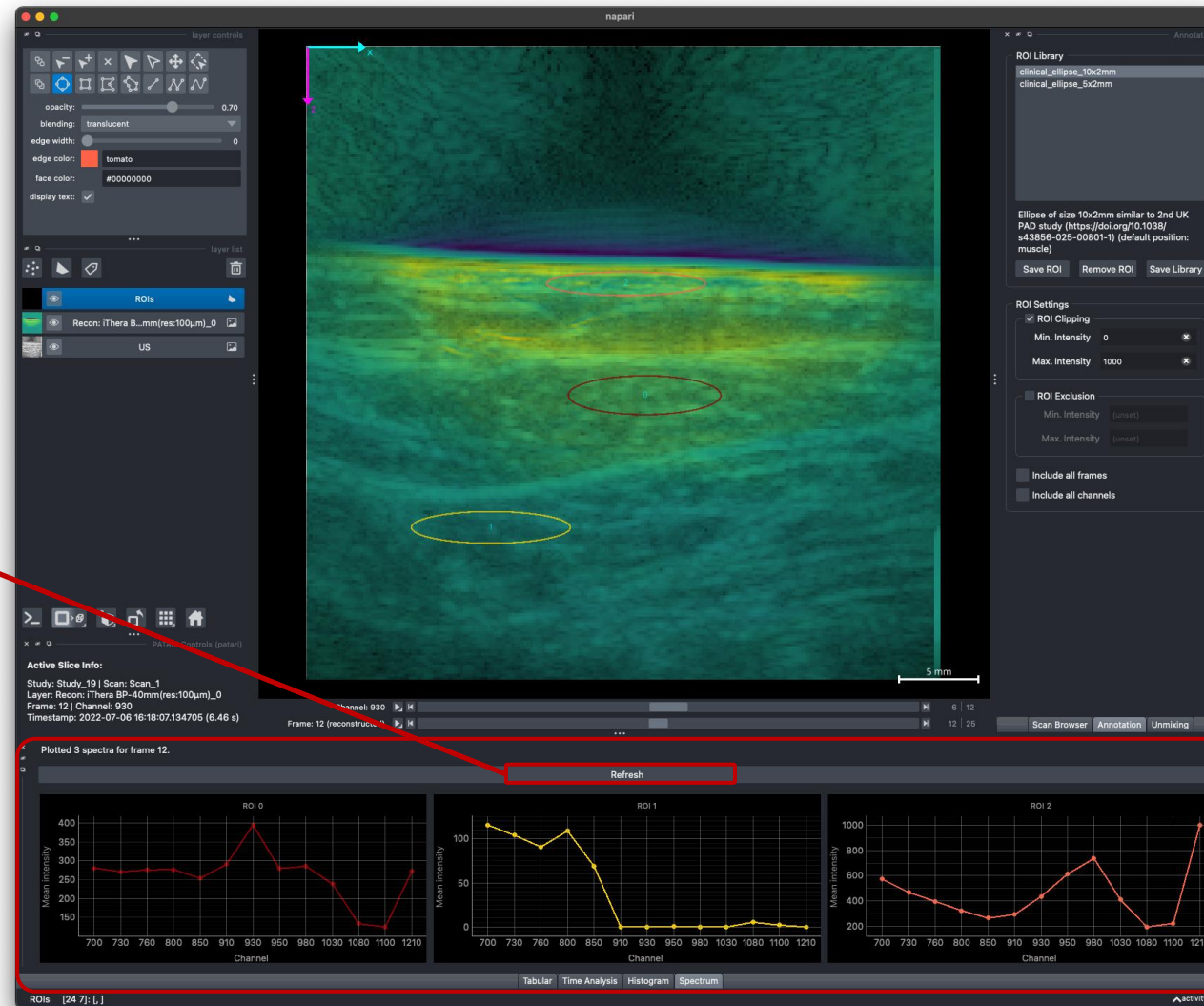
x-axis: if timestamps are available, relative time since the first frame. Otherwise frames.

Histogram Analysis



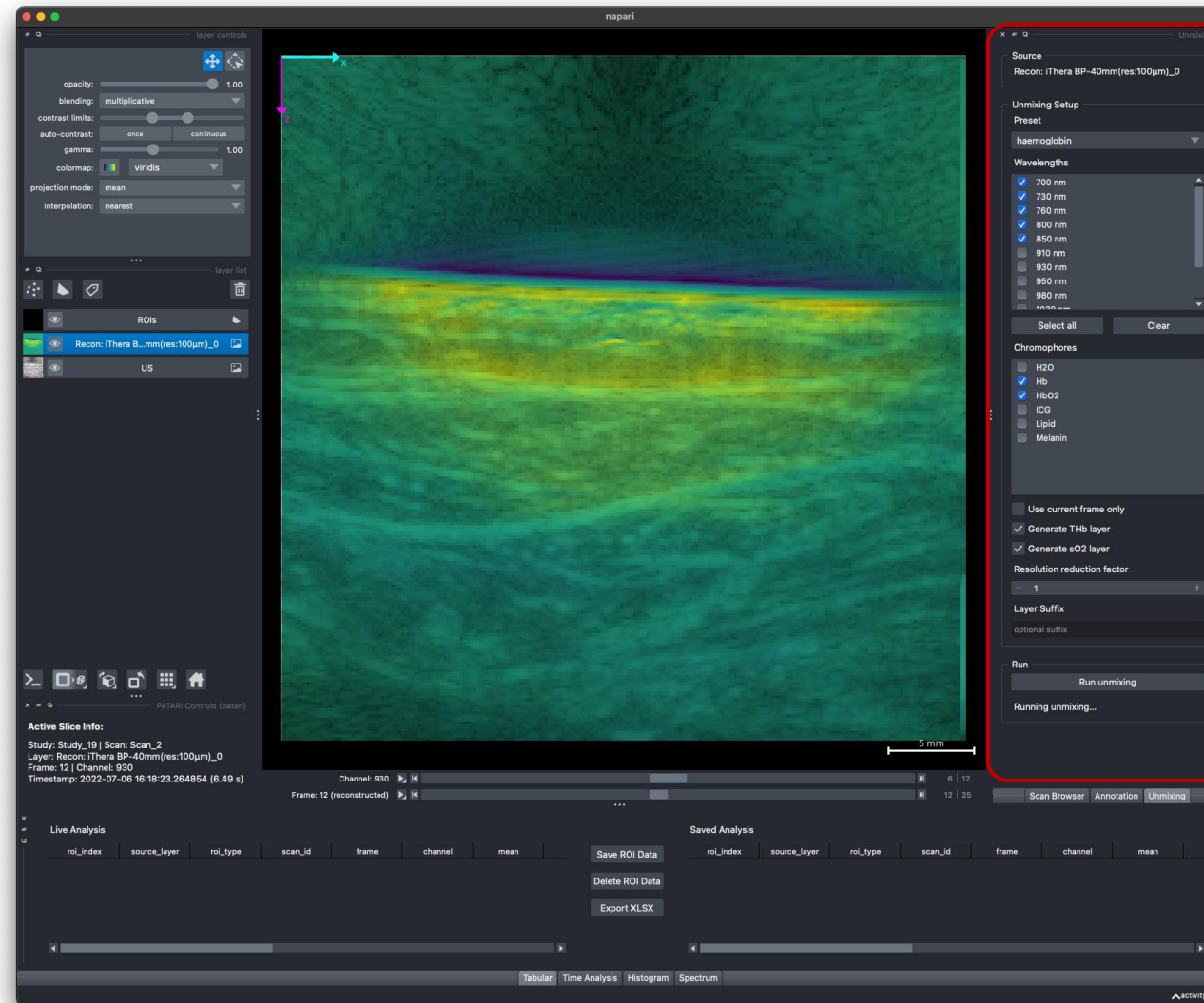
Unlike the Live Table, plots are not auto-updated. When ROIs, layer, frame or channel are changed, **Refresh** has to be clicked again.

Plots a histogram for each ROI, showing the distribution of intensities



Unlike the Live Table, plots are not auto-updated. When ROIs, layer or frame are changed, **Refresh** has to be clicked again.

Plots a spectrum for each ROI, showing the intensities in the ROI across the different channels (wavelengths or chromophores)



Unmixing Widget

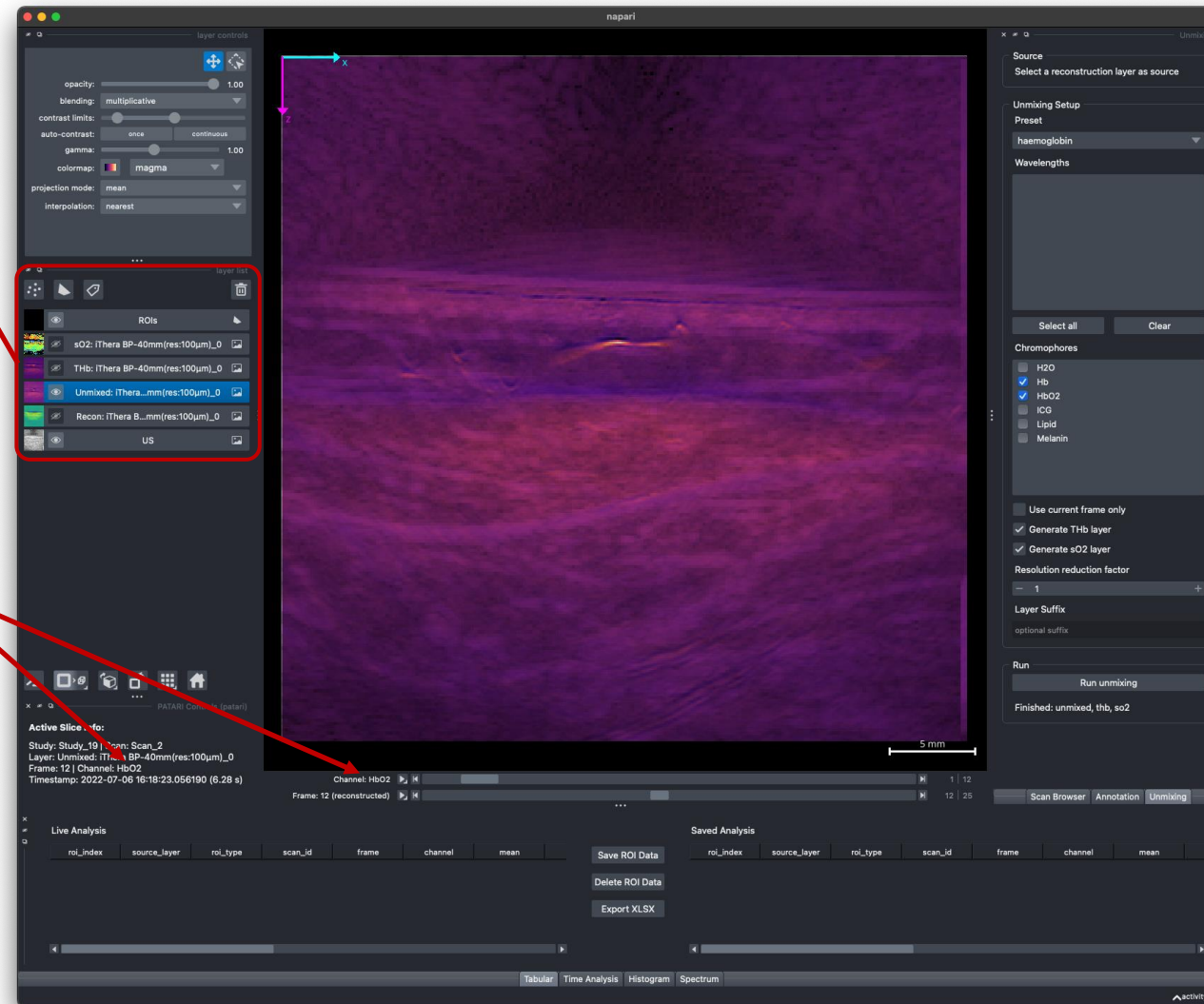
1. Unmixing is performed on the active layer. The active layer must be a PA reconstruction with multiple wavelengths.
2. Select wavelengths to use and target chromophores.
Alternatively, use one of the given presets. Presets can be defined and shared using json files.
3. Select whether all frames of the active layer or just the currently selected frame should be unmixed.
4. Select whether total hemoglobin (THb) and oxygenation (sO2) should also be calculated (requires Hb and HbO2 selection).
5. Choose a resolution reduction factor (1 = no reduction).
6. Add a custom suffix that will be added to the unmixed layers name.
7. Click Run unmixing (can take a few seconds, depending on number of wavelengths and image dimensions).

Unmixing – Layers

Unmixed Layers

Unmixed scans are added as additional layers to the layer list, and can be selected as active layers. If selected, THb and sO2 layers are added as well.

Unmixed layers have chromophores as channel dimension



Example Workflow

Evaluation of a clinical PA study

Example Workflow

Demo Video

1. Open Study folder (first scan is selected automatically)
2. Scroll to target channel and frame using the sliders
3. Select the ROI layer and draw the desired ROI shape that should be used for the analysis
4. Save the ROI to the ROI library with a given name and description
5. Store the ROI data in the Saved Table by clicking Save ROI Data
6. Continue to the next scan, scroll to target channel and frame
7. Place the ROI by double clicking on its name in the ROI Library
8. Move the ROI to the correct location
9. Store the ROI data in the Saved Table by clicking Save ROI Data
10. Continue for all scans, open new study folder, repeat ...
11. Finally, export all the saved ROI data to xlsx by clicking on the Export XLSX button
12. Optional: permanently add the ROI to the ROI Library by clicking Save ROI Library

